WanderGuide: Personalised Planning App

Dakshin Karthick S	**23Z312**
Dinesh T M	**23Z320**
Harish Karthic R	**23Z324**
Naveen Sundar M	**23Z349**
Raja Shylesh S	**23Z356**
Ramnaren N	**23Z357**

23Z515 AICTE ACTIVITY POINT PROGRAMME

Report submitted in partial fulfilment of the requirement for the award of degree of

BACHELOR OF ENGINEERING

Branch: COMPUTER SCIENCE AND ENGINEERING

Of Anna University

Nov 2025

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PSG COLLEGE OF TECHNOLOGY

(Autonomous Institution)

COIMBATORE – 641 004

PSG COLLEGE OF TECHNOLOGY

(Autonomous Institution)

COIMBATORE – 641 004

WANDERGUIDE: AN INNOVATIVE ITINERARY DESIGNING APP

Bonafide record of work done by

Dakshin Karthick S	**23Z312**
Dinesh T M	**23Z320**
Harish Karthic R	**23Z324**
Naveen Sundar M	**23Z349**
Raja Shylesh S	**23Z356**
Ramnaren N	**23Z357**

Dissertation submitted in partial fulfilment of the requirements for the degree of

BACHELOR OF ENGINEERING

Branch: COMPUTER SCIENCE AND ENGINEERING

Of Anna University

Nov 2025

……	……
Dr. V. Santhi	**Dr. G. Sudha Sadasivam**
Programme Coordinator	**Head of the Department**

The candidate was examined in the review held on …….....

……..	……..	……..
Panel Member 1	**Panel Member 2**	**Panel Member 3**
(Dr. V. Santhi)	**(Dr. KM Kirthika)**	**(Dr. S. Sivaranjani)**

ACKNOWLEDGEMENT

The AICTE Activity Point Programme project work had paved the way for personal development and creativity as proud volunteers with a sense of achievement and ready to take up projects having a social impact and creating digital awareness. We feel privileged to be associated with such a curriculum in **PSG College of Technology**.

We are extremely grateful to our **Principal Dr. K. Prakasan** for facilitating and extending untiring support in successful completion of the project. We are extremely glad to extend our gratitude to the **department Head Dr. G. Sudha Sadasivam** and **Programme Co-Ordinator Dr. V. Santhi** for offering us this opportunity which has made us to strengthen our soft skills, leadership qualities and team spirit. We extend our thanks to **Department Placement Co-Ordinator Dr. K Sathiya Priya**, who has been the pillar of support.

We would like to take this opportunity to place our sincere thanks to our **Tutor Dr. KM Krithika** for her guidance in our project work and would also like to thank our **Panel members** who spent their valuable time to aid us for coming out with the good project.

CERTIFICATE

Certified that this report titled "**WanderGuide: An Innovative Itinerary Designing App**" the **AICTE ACTIVITY POINT PROGRAMME** project work (23Z515) is a bona fide work of

"**DAKSHIN KARTHICK S (23Z312), DINESH T M (23Z320), HARISH KARTHIC R (23Z324), NAVEEN SUNDAR M (23Z349), RAJA SHYLESH S (23Z356), RAMNAREN N (23Z357)**"

who have carried out the work for the partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering.

Place : Coimbatore
Date : 06/11/2025

Tutor
(Dr.KM Kirthika)

Department Placement Coordinator
(Dr. K SATHIYA PRIYA)

COUNTERSIGNED

Dr. G Sudha Sadasivam
HEAD
Department of Computer Science and Engineering

PSG College of Technology, Coimbatore – 641 004

CONTENTS

Chapter **Page No.**

Abstract…….. 5

1.	**INTRODUCTION**		**6**
	1.1	Introduction	
	1.2	Problem Statement	
2.	**ACTIVITY PLAN**		**7**
	2.1	Literature Survey	
	2.2	Proposed Plan	
	2.3	Estimated Time	
3.	**SYSTEM DESIGN**		**21**
	3.1	System Requirements	
	3.2	Module Description	
	3.3	System Diagrams	
4.	**IMPLEMENTATION**		**32**
5.	**DISCUSSION**		**30**
	5.1	Discussion on the Proposed Activity	
6.	**ACTIVITY LOG**		**33**

ABSTRACT

Tourism plays a pivotal role in the advancement and promotion of a nation, contributing significantly to its visibility and appeal. However, countries are currently confronted with the challenge of effectively promoting their tourism offerings to both domestic and international audiences. To bridge this gap, many nations are leveraging user-friendly technological tools to enhance awareness of their tourism destinations.

Recognising the immense potential of India's tourism sector and the absence of a comprehensive platform facilitating connections between tourists and destinations, we have embarked on a project to address this need. Our platform will prioritise user-friendly navigation, enabling seamless exploration of various destinations across India. It will provide detailed information on popular tourist spots, encompassing historical landmarks, natural wonders, cultural attractions, and lesser-known locales.

Furthermore, our platform will offer customisation options such as trip duration, hotel preferences, and budget constraints, empowering users to craft personalised itineraries effortlessly through an intuitive drag-and-drop interface. Additionally, we will incorporate a forum feature to foster connections among like-minded travellers, facilitating the exchange of experiences and insights on various tourism-related topics.

Through our initiative, we aim to stimulate economic growth in India's tourism sector while also promoting greater socio-economic equity across different regions of the country.

CHAPTER 1
INTRODUCTION

1.1 Introduction

Tourism significantly boosts a nation's economy and cultural exchange, and India, with its rich heritage and diverse landscapes, holds vast potential in this sector. However, the country struggles to effectively promote its many attractions. To address this, we are developing a user-friendly platform that offers detailed information on various tourist destinations, including popular sites and hidden gems. Features like customisable travel itineraries and a traveller forum will enhance user experience and foster a supportive community. This initiative aims to drive economic growth and promote socio-economic equity by increasing the visibility and accessibility of India's diverse regions.

1.2 Problem Description

Despite India's rich cultural heritage, diverse landscapes, and numerous tourist attractions, the country faces significant challenges in effectively promoting its tourism offerings to both domestic and international audiences. The lack of a comprehensive, user-friendly platform that connects tourists with the vast array of destinations across India results in underutilisation of the sector's potential. This gap hinders the economic growth of the tourism industry and limits socio-economic equity across different regions.

There is a pressing need for an innovative solution that provides seamless navigation, detailed information, and customisable travel itineraries, enabling tourists to explore India's popular and lesser-known destinations with ease. Furthermore, fostering a community where travellers can share experiences and insights can enhance the overall tourism experience.

Our project aims to develop a comprehensive platform that addresses these needs, enhancing the promotion and accessibility of India's tourism destinations, stimulating economic growth, and promoting greater socio-economic equity across the country.

CHAPTER 2
ACTIVITY PLAN

2.1 Literature Survey

2.1.1 TripAdvisor's Trip Builder

TripAdvisor's Trip Builder is an innovative online tool designed to simplify the travel planning process. With a user-friendly interface and a wealth of resources, Trip Builder allows travellers to customise their itineraries, ensuring a personalised experience tailored to their interests and preferences. Whether you're a solo adventurer, a family on vacation, or a couple looking for a romantic getaway, Trip Builder provides the tools needed to create unforgettable travel experiences.

Key Features

1. **Customisable Itineraries:**

   Easily plan your trip by adding attractions, restaurants, and activities that match your interests and schedule.

2. **Interactive Map:**

   Visualise your travel plans with an integrated map that helps you organise your itinerary based on location, making it easier to navigate your destination.

3. **Recommendations and Reviews:**

   Access a vast database of user-generated reviews and ratings to make informed decisions about where to go and what to do.

4. **Collaborative Planning:**

   Share your trip plans with friends and family, allowing for collaborative input and feedback on the itinerary.

5. **Budgeting Tools:**

   Estimate costs for your trip by viewing pricing information for accommodations, activities, and dining options, helping you stay within your budget.

Advantages

1. **Enhanced Personalisation through AI Recommendations:**

   Trip Builder's AI-driven algorithms analyse user preferences, past behaviours, and similar traveller profiles to suggest personalised recommendations. This allows users to see curated options based on interests, budget, and travel style, such as

solo or family travel, enhancing the relevance of suggestions and creating a tailored travel plan.

2. **Dynamic Adaptability for Real-Time Customisation:**

   Leveraging AI, Trip Builder offers real-time itinerary adjustments based on changing preferences or circumstances, such as weather changes or unexpected closures. Users can adjust their plans dynamically, and the platform will generate alternate suggestions or time allocations, ensuring a smooth and flexible travel experience even on short notice.

3. **Intelligent Recommendations Based on Location and Interests:**

   With AI, Trip Builder can provide location-based suggestions that align with user interests, such as nearby dining spots, hidden attractions, or unique cultural experiences that other users with similar tastes have enjoyed. This feature helps travellers uncover hidden gems tailored specifically to their interests, which can enhance the overall travel experience.

4. **Time-Efficient Trip Planning through Streamlined, AI-Filtered Options:**

   AI-powered sorting and filtering save users time by showing them the most relevant options upfront. Rather than sorting through an overwhelming amount of content, users receive targeted suggestions based on their preferences, making the planning process faster and more efficient.

5. **Consistent Refinement of Recommendations via Machine Learning:**

   Trip Builder learns from each user's engagement with the platform, continuously refining its recommendations over time. For frequent travellers, the platform becomes increasingly accurate, presenting a progressively better-suited travel experience each time it's used. This iterative improvement supports loyalty and helps the platform become an increasingly reliable travel companion.

Disadvantages

1. **Potential for Algorithmic Bias in Recommendations:**

   While AI can improve personalisation, it can also create biases in recommendations based on past preferences or highly rated/popular spots, potentially limiting users' exposure to new experiences. This bias may cause the platform to over-prioritise familiar or mainstream destinations rather than suggesting unique or off-the-beaten-path locations.

2. **Privacy Concerns with Data Usage and Personalisation:**

   To deliver personalised experiences, the AI algorithms analyse personal data, such as past bookings, browsing behaviour, and demographic information. While this

enables personalisation, it also raises privacy concerns, as users may feel uneasy about the extent of data collection needed to provide tailored recommendations.

3. **Dependence on Data Quality for AI Effectiveness:**

   The AI's ability to deliver accurate recommendations depends heavily on the quality and breadth of available data. In cases where there is limited information on newer or lesser-known destinations, recommendations may be less accurate or overly reliant on popular spots, which could diminish the user experience for those seeking unique travel options.

4. **Risk of Narrowing Choices Due to Personalisation:**

   Personalised suggestions may limit the variety of options shown, inadvertently filtering out activities or destinations that could have been a good fit outside of previous patterns. This "filter bubble" effect might prevent users from exploring novel experiences that differ from their past interests, leading to a more homogenised travel experience.

5. **Over-Reliance on AI for Travel Decision-Making:**

   The AI-driven interface may reduce users' autonomy by heavily influencing their choices based on pre-set algorithms. While this can simplify planning, it might also lead users to depend on the platform's recommendations over personal research, potentially missing out on richer experiences that require a more proactive, hands-on approach to discovery.

2.1.2 getyourguide.com

Introduction

GetYourGuide is a popular travel platform that allows users to discover and book a diverse range of activities and experiences in various destinations. With its extensive catalogue and personalised recommendations, the app seeks to cater to the unique interests of travellers. This paper explores the key features of GetYourGuide, its benefits, and potential drawbacks.

Key Features

1. **Extensive Activity Catalogue:**

   GetYourGuide offers thousands of activities across the globe, including guided tours, cultural experiences, outdoor adventures, and culinary classes. This extensive selection ensures travellers can find activities that suit their interests, whether they're seeking thrill or relaxation.

2. **Personalised Recommendations:**

   The app analyses user data, including past bookings and preferences, to provide tailored activity suggestions. This feature helps users discover unique experiences that align with their tastes, making the travel planning process more efficient and enjoyable.

3. **User Reviews and Ratings:**

   GetYourGuide includes a robust review system where travellers can read detailed feedback from others who have participated in the activities. This transparency helps users make informed decisions and enhances trust in the platform, as they can gauge the quality of experiences before booking.

4. **Secure Booking and Payment:**

   The app priorities user security by offering secure payment options and a streamlined booking process. Many activities also come with flexible cancellation policies, providing users with peace of mind and the ability to adjust their plans without financial penalty.

5. **Last-Minute Booking Capability:**

   GetYourGuide allows users to book activities at the last minute, which is ideal for spontaneous travellers or those who prefer to plan on the go. This flexibility ensures that even last-minute travellers can enjoy curated experiences without extensive prior planning.

Advantages

1. **Wide Range of Options:**

   The diverse catalogue allows travellers to find activities that cater to various interests and age groups, enriching their travel experience. Whether users are interested in historical tours or adventurous outings, GetYourGuide provides ample choices to create a customised itinerary.

2. **User-Friendly Interface:**

   The app's intuitive design makes it accessible to all users, regardless of their tech-savviness. Easy navigation, clear visuals, and straightforward booking processes enhance user satisfaction, making travel planning less daunting.

3. **Convenience and Timesaving:**

   By consolidating multiple travel services into one platform, GetYourGuide saves users time and effort in planning their trips. Instead of browsing multiple sites, travellers can quickly find and book activities all in one place, streamlining the overall planning process.

4. **Real-Time Updates:**

   Users receive instant notifications about their bookings, including any changes or updates, allowing them to stay organised. This feature helps travellers manage their itineraries effectively, reducing the risk of missed opportunities or double bookings.

Disadvantages

1. **Pricing Variability:**

   Although GetYourGuide often provides competitive pricing, there may be instances where direct bookings with local providers offer lower rates. This variability can lead to frustration for users who are budget-conscious and looking for the best deals.

2. **Dependence on Internet Access:**

   As a mobile app, GetYourGuide requires a stable internet connection to function properly. Travellers in remote locations with limited connectivity may find it difficult to access their bookings or search for activities, which could disrupt their plans.

3. **Overwhelming Choices:**

   While having numerous options is generally beneficial, it can also lead to decision fatigue. Users may feel overwhelmed by the plethora of activities available, making it challenging to select the best options for their needs and preferences.

4. **Variable Quality of Experiences:**

   The quality of activities offered on GetYourGuide can differ significantly based on the provider. Despite user reviews, some travellers may encounter experiences that do not meet their expectations, impacting their overall satisfaction with the app.

2.1.3 Booking.com: The biggest hotel booking service in the world

Booking.com: Agency Model vs. Merchant Model

Booking.com primarily uses the agency model, which differs from the merchant model used by some other online travel agencies (OTAs) like Expedia:

Agency Model: In this model, Booking.com serves as an intermediary, connecting customers with hotels. Hotels control the room rates and availability while Booking.com earns a commission on each booking. Payment is generally made directly to the hotel upon check-in or check-out, giving the hotel control over pricing and customer service. This approach also allows travellers the option to pay later and typically includes more flexible cancellation options.

Merchant Model: In contrast, the merchant model involves the OTA purchasing rooms in bulk at a discounted rate, then reselling them to customers with a markup. Payment is made upfront to the OTA, rather than the hotel, which can offer competitive pricing but often involves less flexibility in cancellation or refund options.

Booking.com's choice of the agency model prioritises flexibility and direct customer relationships, particularly beneficial for travellers seeking adaptable bookings.

Key Features

1. **Wide Inventory:**

   Booking.com has a vast range of accommodations worldwide, including hotels, vacation rentals, apartments, and unique options like treehouses.

2. **Flexible Cancellation Policies:**

   Many listings offer options for free cancellation, appealing to travellers needing flexibility.

3. **User Reviews and Ratings:**

   Verified reviews help customers make informed choices based on real guest experiences.

4. **Genius Loyalty Program:**

   Frequent users benefit from exclusive discounts, room upgrades, and perks like complimentary breakfast.

5. **Price Match Guarantee:**

   Booking.com matches lower prices found elsewhere to ensure competitive pricing.

6. **Mobile App Convenience:**

   The app allows easy booking management, personalised recommendations, and quick access to reservation details.

Advantages of Booking.com Using an Agency Model

1. **Low Operational Risk for Booking.com:**

   Booking.com does not need to purchase inventory upfront. They act as a mediator, reducing risks associated with unsold inventory.

2. **Flexibility for Partners (Hotels/Hosts):**

   Hotels and accommodation providers retain control over pricing, availability, and policies, making the model more appealing to businesses.

3. **Scalable Business Model:**

   Booking.com can easily onboard more partners without needing to invest in inventory, enabling rapid expansion into new markets.

4. **Simplified Revenue Stream:**

   The commission-based revenue is straightforward and predictable, as Booking.com earns a percentage from each successful booking.

5. **Improved Customer Trust:**

   Customers pay the accommodation provider directly, which can enhance trust and confidence in their transactions.

Disadvantages of Booking.com Using an Agency Model

1. **Dependency on Partners:**

   Booking.com depends on its partners to provide quality service, availability, and pricing, which can impact the customer experience if not managed well.

2. **Limited Control Over Pricing:**

   Since partners set their own prices, Booking.com has limited influence on pricing strategies, which could make it harder to remain competitive.

3. **Delayed Revenue Collection:**

   Booking.com earns its commission only after the customer completes their stay, delaying revenue realisation compared to the upfront payments in a merchant model.

4. **Lower Margins:**

   Compared to the merchant model, where platforms can mark up prices, the agency model has smaller profit margins due to reliance on a fixed commission structure.

5. **Risk of Cancellation:**

   If customers cancel bookings, Booking.com loses its commission revenue, which could affect earnings.

6. **Increased Competition:**

   The same model is used by many competitors, leading to intense competition in attracting both customers and partners.

2.1.4 cleartrip.com

Cleartrip offers a robust and user-friendly travel planning experience, aimed at simplifying trip arrangements with a seamless blend of booking options, itinerary management, and personalised recommendations. With features designed for both novice and seasoned travellers, Cleartrip provides an all-in-one platform to book flights, hotels, and activities. Its design emphasises ease of use, encouraging smooth navigation and planning.

Key Features

1. **Comprehensive Booking Platform:**

   Cleartrip enables travellers to book flights, hotels, and activities from a centralised interface, ensuring a cohesive planning process across multiple categories.

2. **Flexible Itinerary Management:**

   Users can manage itineraries by adjusting bookings, adding activities, and making last-minute changes without the need to switch between multiple apps or websites.

3. **Personalised Recommendations:**

   Cleartrip uses AI to recommend flights, accommodations, and activities that align with user preferences, past bookings, and travel behaviour.

4. **Price Alerts and Budget Tools:**

   The platform offers automated alerts for fare drops, deals, and special discounts, allowing users to plan their trips without exceeding their budgets.

5. **Multi-Destination Trips:**

   Users can build multi-city itineraries by adding multiple destinations within a single trip, streamlining complex travel plans across several locations.

6. **Collaborative Planning:**

   Travellers can share itineraries with friends or family, enabling collaborative decision-making and input for group trips.

Advantages

1. **AI-Driven Personalisation for Enhanced Experiences:**

   Cleartrip's algorithms suggest flights, hotels, and activities based on past preferences, behaviour, and trip history. This ensures highly relevant and personalised travel recommendations, enhancing the user experience.

2. **Dynamic Adaptability for Last-Minute Adjustments:**

   The platform allows users to make changes to their plans, such as rescheduling flights or switching accommodations, without hassle. Real-time updates ensure smooth rebooking options based on cancellations or delays.

3. **Streamlined Navigation and Booking Across Categories:**

   By consolidating flights, hotels, and activities in one interface, Cleartrip eliminates the need to browse multiple services, making the planning experience faster and more efficient.

4. **Location-Based Suggestions and Hidden Gems:**

   Cleartrip's AI recommends attractions, dining options, and offbeat experiences based on a traveller's location and preferences. This feature helps users uncover unique activities that align with their interests, enhancing the trip's quality.

5. **Budget Tracking and Alerts for Cost-Efficiency:**

   Cleartrip's budget tools and alerts help users keep track of travel expenses, offering alternatives to stay within budget through deals and personalised discounts.

Disadvantages

1. **Algorithmic Bias Towards Popular Choices:**

   While Cleartrip offers personalised recommendations, it may overemphasise well-known destinations and accommodations, potentially reducing exposure to lesser-known or offbeat travel options.

2. **Privacy Concerns with Data Usage:**

   To deliver personalised services, Cleartrip collects and analyses user data, which might raise privacy concerns regarding how personal information is stored and used.

3. **Reliance on Data Quality for Accuracy:**

   The AI recommendations depend on the availability of data. For newer or niche destinations, the suggestions might not be as accurate or insightful, potentially limiting the user's experience.

4. **Risk of Over-Personalisation Limiting Variety:**

   Personalised recommendations may unintentionally narrow the range of travel options shown, filtering out experiences outside of the user's past preferences, leading to repetitive travel plans.

5. **Reduced Autonomy Due to Heavy AI Influence:**

   Over-reliance on Cleartrip's AI-powered recommendations may limit users from conducting independent research. This could discourage travellers from discovering unique activities that fall outside the algorithm's suggestions.

2.1.5 Relevant Research

Title: *Smart Destinations and the Evolution of ICTs: A New Scenario for Destination Management?*

Authors: Josep A. Ivars-Baidal, Marco A. Celdrán-Bernabeu, Jose-Norberto Mazón, Ángel F. Perles-Ivars

Publication: *Current Issues in Tourism, October 2017*

Abstract

This paper examines the transformative impact of information and communication technologies (ICTs) on tourism destination management, proposing a smart tourism destination (STD) model that integrates strategic, instrumental, and applied technological levels. Using the Delphi technique, the authors gather expert insights into the STD approach's feasibility, advantages, and limitations. The findings suggest that, while emerging ICTs such as big data and IoT enhance data-driven decision-making and visitor experience, successful implementation hinges more on effective governance and stakeholder cooperation than technology alone. The study highlights the short-term importance of these technologies, projecting significant influence on destination management within 2–5 years.

Key Observations

- **The Role of ICT in Tourism Management:**

  ICTs provide real-time data capabilities that transform the tourism experience, allowing destinations to better meet visitor expectations through personalized services. This data-driven approach aligns with the trend towards experience-driven tourism and supports sustainability efforts by enabling more efficient resource management.

- **Governance as a Core Challenge:**

  The success of STDs depends largely on robust governance frameworks that facilitate cooperation among stakeholders. Technology alone is insufficient; collaborative, strategic governance is essential to address barriers like interoperability, high costs, and limited public-private collaboration.

- **Emerging Technology Impact:**

  Big data analytics, IoT, and mobile connectivity are identified as high-impact technologies, expected to transform tourism within a short term (2–5 years). Technologies such as augmented reality and autonomous vehicles are anticipated to mature later but are forecasted to play significant roles in enhancing the tourist experience and operational efficiencies.

Advantages

1. **Enhanced Tourist Experiences:**

   The STD model enables destinations to deliver personalised, real-time tourist interactions, enhancing visitor satisfaction and creating memorable experiences. This allows destinations to stand out by tailoring experiences to individual preferences, leveraging big data and analytics.

2. **Competitive Edge and Sustainability:**

   STDs improve the operational efficiency of destinations, offering an advantage over traditional models. By integrating ICTs, destinations can optimise resource management, especially in ecologically sensitive areas, and use real-time data to respond dynamically to visitor needs.

3. **Increased Collaboration and Innovation:**
   The model fosters collaboration among tourism operators, technology providers, and local governments, driving innovation and supporting new product development that aligns with evolving tourist expectations.

Disadvantages

1. **High Implementation Costs:**

   Implementing ICT-driven destination management systems requires significant upfront investment, which can be challenging for smaller destinations with limited budgets. Continuous updates and maintenance add to the financial burden, potentially limiting accessibility to this model.

2. **Privacy and Security Concerns:**

   As STDs depend heavily on data collection, they raise privacy and cybersecurity issues. Safeguarding visitor data is critical to maintaining trust, yet costly cybersecurity measures and compliance with data protection regulations are additional challenges for destination managers.

3. **Technological Dependency and Skills Gap:**

   The reliance on advanced technology can reduce human interaction, which may be a preference for some tourists. Additionally, the rapid pace of technological

change demands continuous staff training, which can be challenging, particularly in tourism sectors with high turnover rates and limited ICT expertise.

2.2 Proposed Plan

1. Project Initiation and Research:

- **Define Project Scope and Objectives:**
  - Focus on personalised itinerary generation without relying on machine learning algorithms.
  - Identify alternative methods for user profiling and recommendations (e.g., rule-based systems, user surveys, explicit preference settings).
  - Research potential data sources for attractions, transportation, accommodations, and local events.
- **Data Acquisition and Validation:**
  - Gather comprehensive and accurate data from reliable sources.
  - Explore partnerships with local businesses and tourism boards for exclusive content and deals.
  - Establish data validation processes to ensure accuracy and relevance.

2. Design and Development:

- **UI/UX Design:**
  - Design user interfaces that facilitate personalised trip planning and itinerary customisation.
  - Develop intuitive features for filtering, sorting, and comparing options.
  - Create a user-friendly experience for providing feedback and preferences.
- **Application Development:**
  - Implement core functionalities for itinerary generation, booking, and search.
  - Integrate with relevant APIs for transportation options, accommodation booking, and local events.
  - Build systems for data processing and recommendations based on user inputs and preferences.

3. Launch and Rollout:

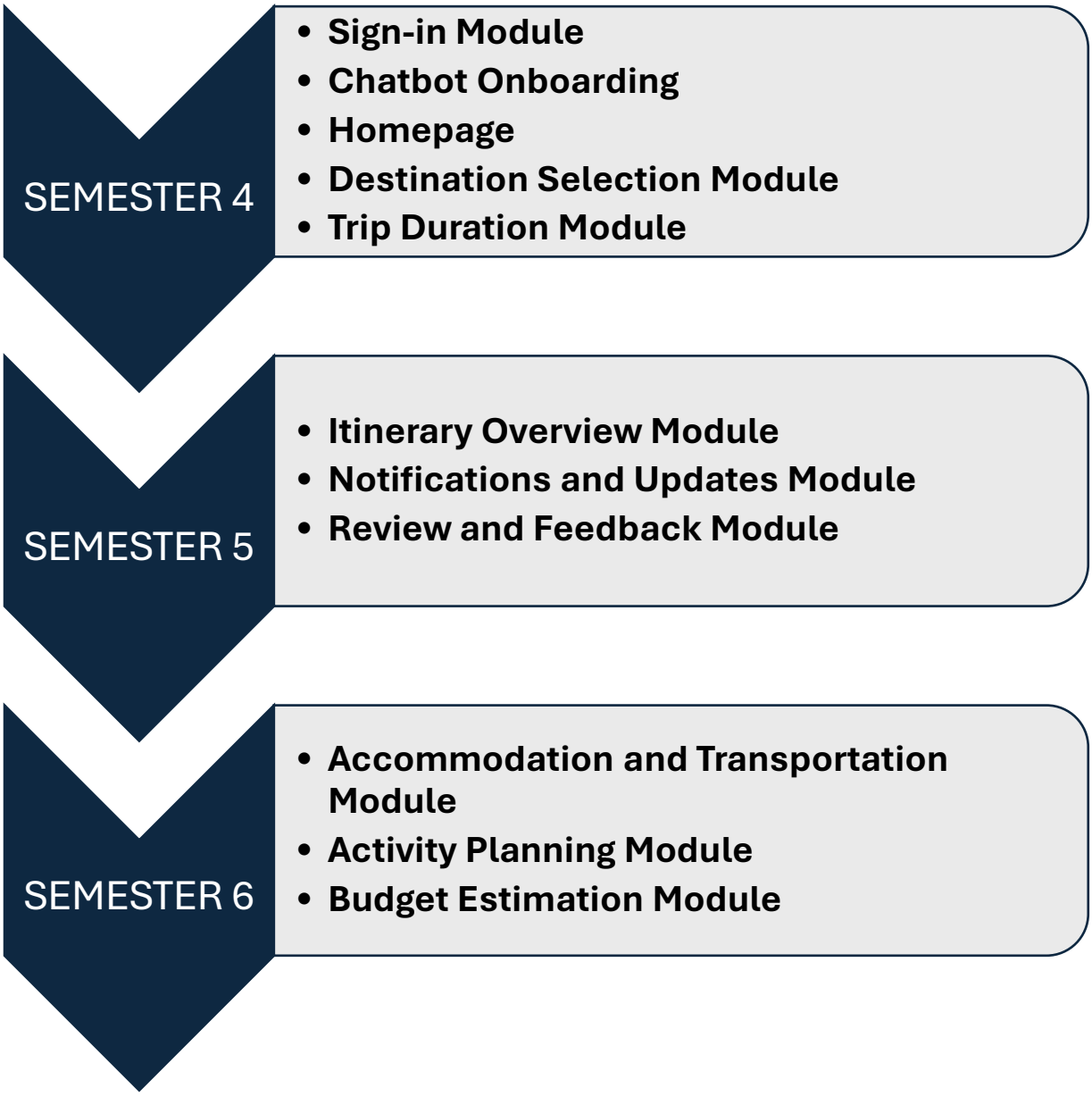
- **Beta Testing and Refinement:**
  - Conduct extensive beta testing with a diverse user group to gather feedback.
  - Iterate on the app based on user insights and preferences.
  - Refine recommendation algorithms based on real-world usage data.

- **App Store Submission:**
  - Prepare app listing with compelling descriptions, screenshots, and promotional videos.
  - Submit the app to relevant app stores (Apple App Store, Google Play Store).
- **Marketing and Promotion:**
  - Launch targeted marketing campaigns on social media, travel websites, and relevant platforms.
  - Partner with travel influencers and bloggers to increase visibility.
  - Offer promotions and discounts to attract early adopters.

4. **Community Engagement and Impact Assessment:**

- **User Feedback and Reviews:**
  - Encourage users to rate, review, and provide feedback through in-app mechanisms.
  - Actively respond to feedback and implement improvements.
- **Impact Assessment:**
  - Track key performance indicators (KPIs) like user engagement, downloads, bookings, and revenue.
  - Analyse user data to understand preferences, behaviour, and identify areas for improvement.
- **Continuous Improvement and Expansion:**
  - Regularly update the app with new features, content, and destinations.
  - Explore partnerships with local businesses to offer exclusive deals and experiences.
  - Expand the app to new markets and adapt to different cultural preferences.

2.3 Estimated Time



CHAPTER 3
SYSTEM DESIGN

3.1 System Requirements

3.1.1 Tools Required

Frontend (User Interface)

Languages:

HTML5: For structuring web pages (e.g., destination information, booking forms).

CSS3: For styling (e.g., responsive designs for mobile and desktop users).

JavaScript: For dynamic interactions like carousels, search filters, and map integrations.

Frameworks/Libraries:

React.js: For building a responsive, component-based UI (e.g., booking system, user profiles).

Next.js: For server-side rendering and better SEO (important for travel-related websites).

Leaflet.js or Google Maps API: For interactive maps to show destinations, routes, and nearby places.

Styling:

Tailwind CSS: For creating a modern, customisable design.

Material-UI: To use pre-built components that follow material design principles.

Features:

Lazy Loading: To ensure quick loading of images (e.g., destination photos).

State Management: React Query or Redux for managing data like flight/hotel availability.

Backend (Logic and Data Handling)

Framework:

Node.js with Express.js: For fast and scalable server-side development.

Database:

PostgreSQL: To store structured data like user accounts, bookings, and reviews.

MongoDB: For flexible storage of destination metadata and user-generated content.

APIs and Integrations:

RESTful API or GraphQL: For communication between the front and back ends.

Travel APIs:

Amadeus API: For flight, hotel, and car rental data.

Skyscanner API: For flight search and pricing.

Booking.com API: For hotel booking data.

Payment Gateway Integration:

Stripe, PayPal, or Razorpay for secure transactions.

Authentication:

Passport.js: For user authentication with email or social logins (Google, Facebook).

JWT (JSON Web Tokens): For secure session handling.

3.2 Module Description

1. **Sign-in Module:** This module ensures secure access to the app, allowing users to sign in using Google or other services. Once signed in, users can save their itineraries, access past trips, and sync their preferences across devices. The sign-in also allows the app to provide personalised suggestions based on the user's previous trips or interests.

2. **Chatbot Onboarding:** The chatbot onboarding module acts as the conversation hub of the app. Users are guided through friendly, step-by-step questions about their travel style, preferred climate, dates, duration, and budget. A smart suggestion flow helps them refine ideas, compare locations, and understand trip feasibility in real time. The interface keeps things simple, with quick replies, voice input, and past preferences auto-filled to speed up planning and remove confusion.

3. **Homepage:** The homepage acts as the control centre of the app. Users are greeted with a dashboard showing quick access to their upcoming trips, saved itineraries, and a clear "Plan New Trip" button. A clean layout provides easy navigation to

explore new destinations, view budget suggestions, or modify an ongoing itinerary.

4. **Destination Selection Module:** Here, users can explore and select destinations for their trip. The app offers intelligent suggestions based on user preferences, trending locations, or even upcoming events (e.g., festivals, concerts). Users can search for places directly, view travel guides, and select multiple locations if they are planning a multi-stop trip.

5. **Trip Duration Module:** Users can set the dates and overall length of their trip. A simple calendar interface lets them pick travel dates, and the app automatically calculates how many days they have at each destination. This feature ensures that the itinerary is perfectly aligned with the trip duration, avoiding overbooking or under-planning.

6. **Accommodation and Transportation Module:** This module helps users find and book the best accommodations and transportation for their trip. The app provides tailored recommendations based on the user's budget and preferences (e.g., luxury hotels, budget hostels, car rentals). It also suggests the most efficient travel options between destinations (flights, trains, or road trips), with price comparisons to help the user make informed decisions.

7. **Activity Planning Module:** Once the destinations are set, users can start planning activities. The app provides curated lists of must-see attractions, events, and experiences, all based on the user's interests. Users can drag and drop activities into their itinerary, with suggested times to make sure each day is well-paced. They can also explore nearby restaurants and nightlife options, ensuring a well-rounded experience.

8. **Budget Estimation Module:** This module gives users a clear idea of how much their trip will cost. It breaks down expenses by accommodation, transportation, food, and activities, allowing the user to adjust based on their preferences. If the total exceeds the user's budget, the app suggests alternatives (e.g., a cheaper hotel or free activities) to bring it back within range. This ensures that users can plan a trip that fits their financial needs.

9. **Itinerary Overview Module:** After adding activities, accommodations, and transportation details, users can view their full itinerary on this page. It presents a clean, structured timeline of the trip, showing each day's plan, activity timings, and travel information. Users can make quick edits, move activities around, or remove them entirely. The itinerary is synced in real-time, so users can access it on their mobile device while traveling.

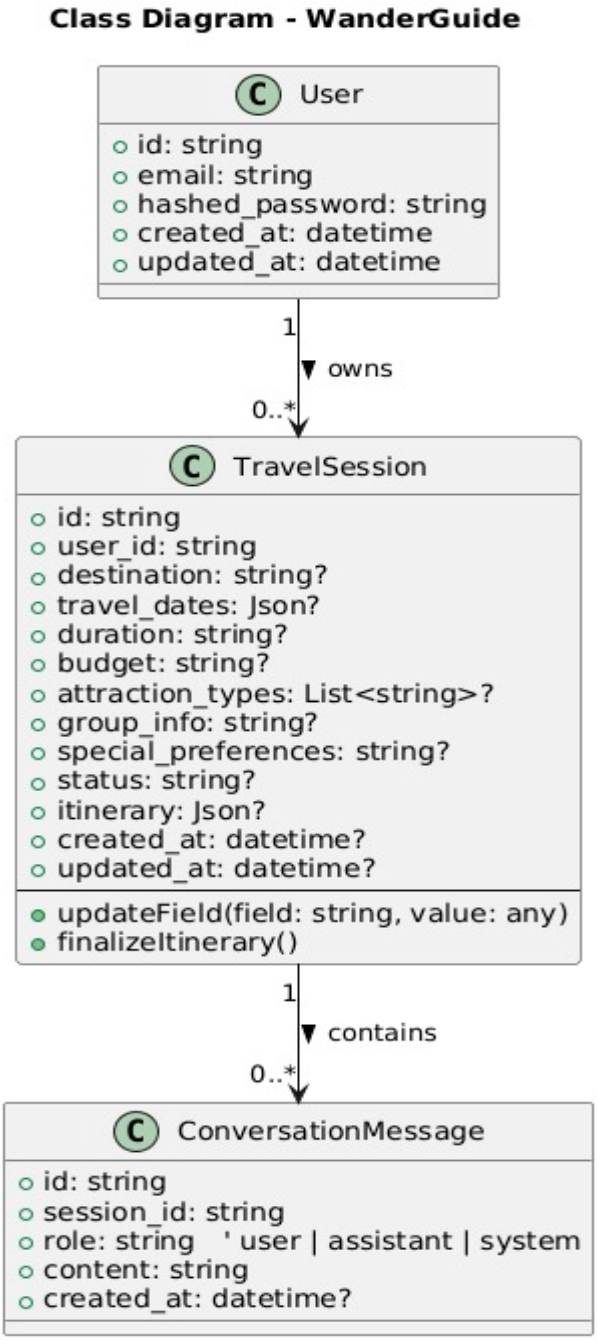
10. **Notifications and Updates Module:** As the trip approaches, users receive real-time notifications about flight changes, weather updates, or even recommendations for

local events happening during their stay. This module ensures that users stay informed about any last-minute changes and can adjust their itinerary accordingly. Push notifications are sent to the user's phone, ensuring they don't miss important updates.

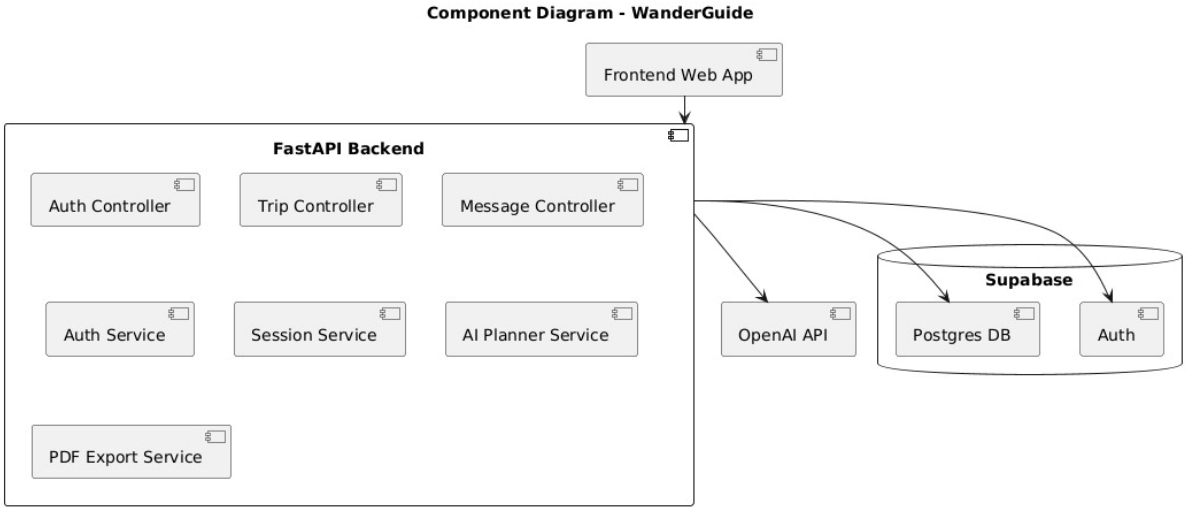
11. **Review and Feedback Module:** Once the trip is complete, the app prompts the user to leave feedback. This module asks for ratings on accommodations, activities, and travel experiences. Users can also write reviews and share their thoughts, which helps the app improve future recommendations. This feedback loop ensures that the app continually evolves based on real user experiences.

3.3 System Diagrams

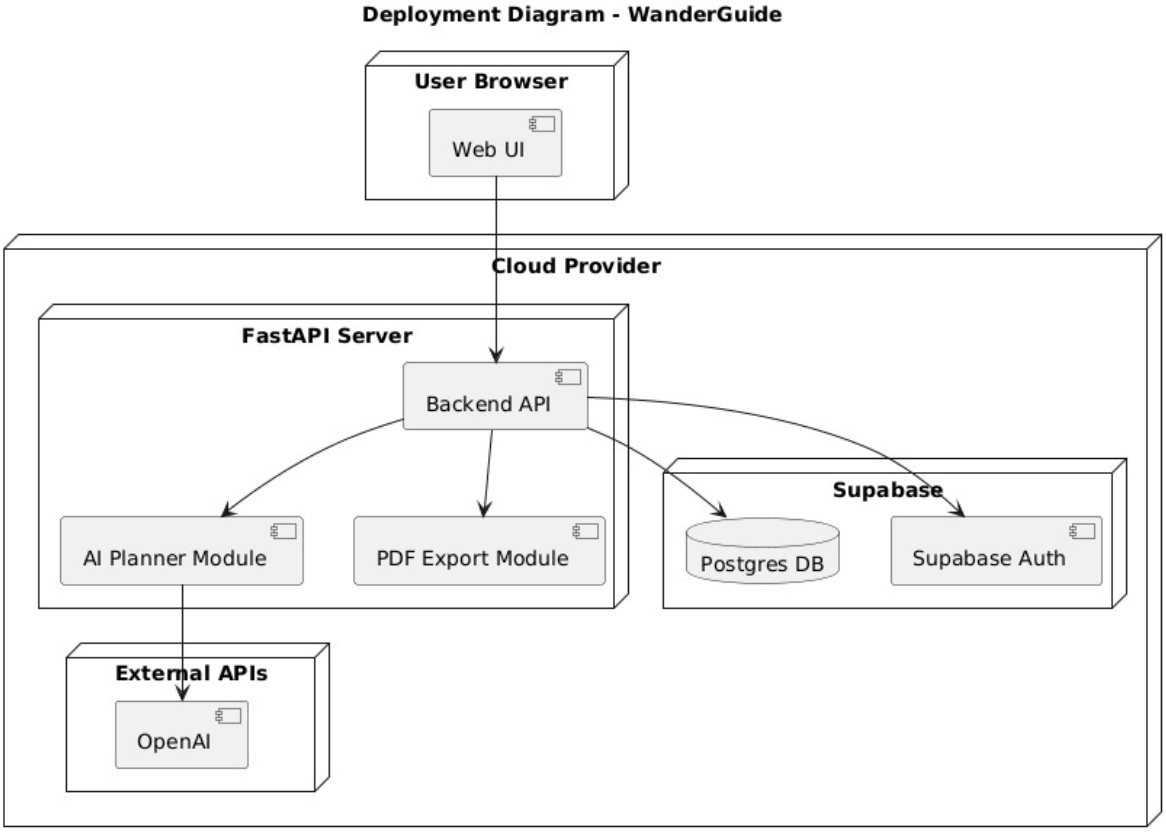
3.3.1. Class Diagram



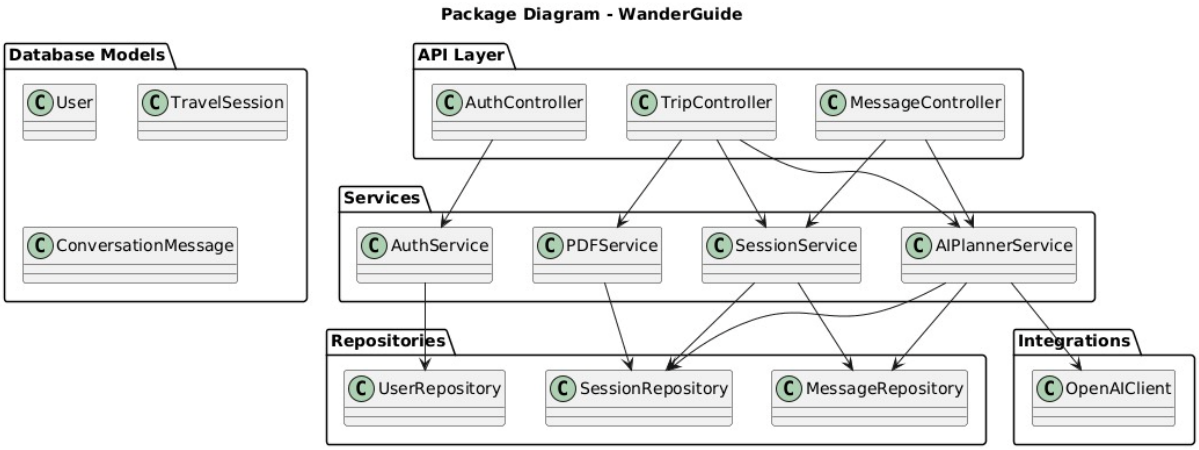
3.3.2 Component Diagram



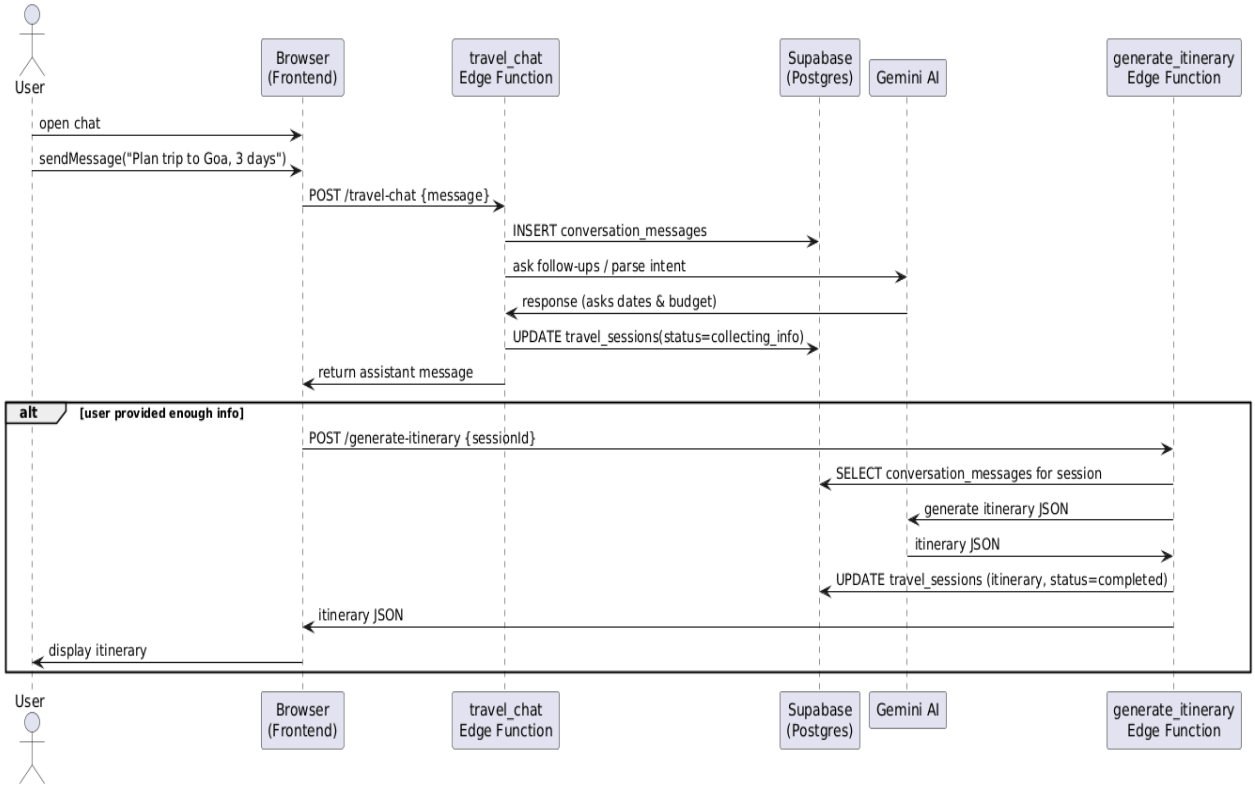
3.3.3 Deployment Diagram



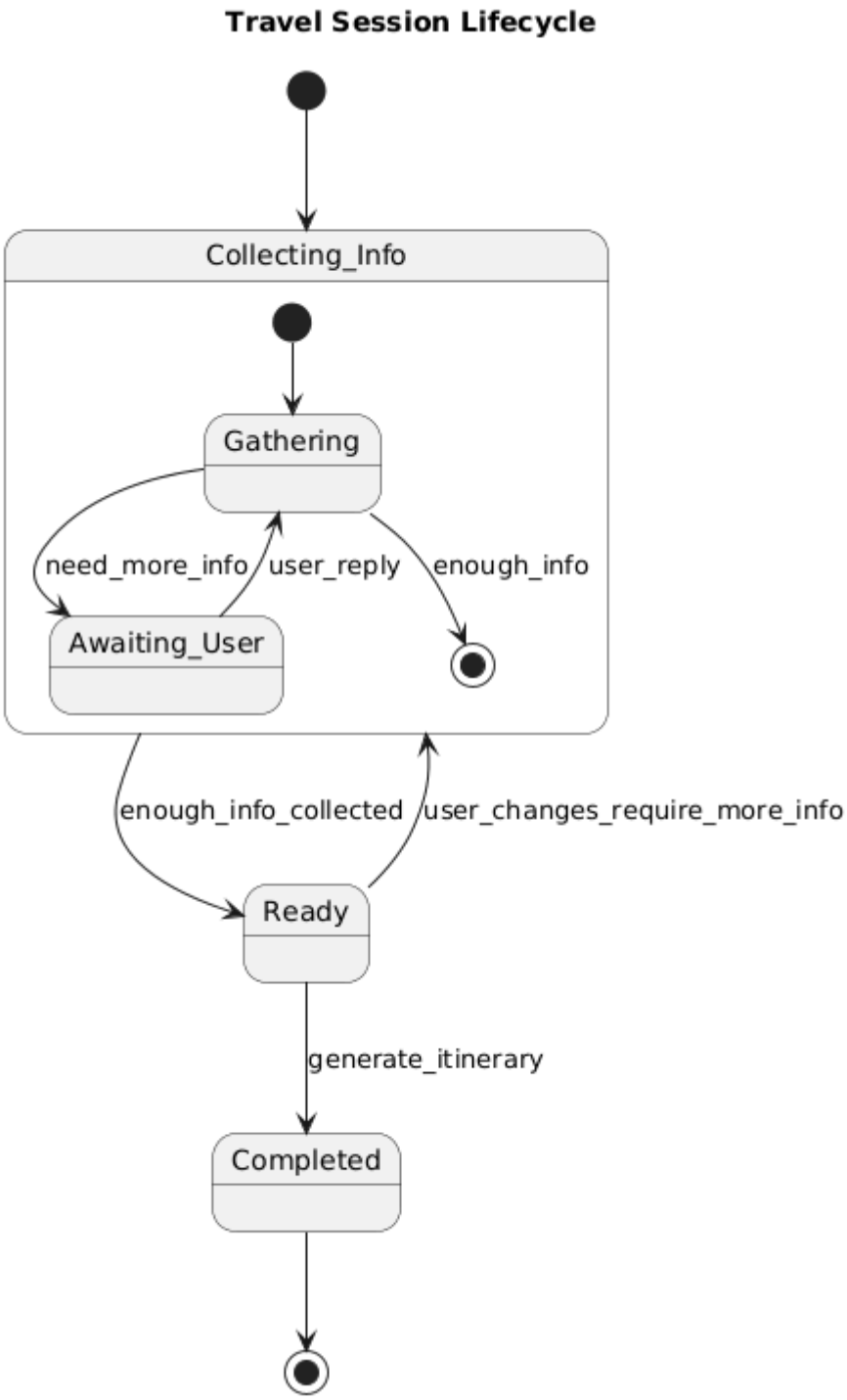
3.3.4 Package Diagram



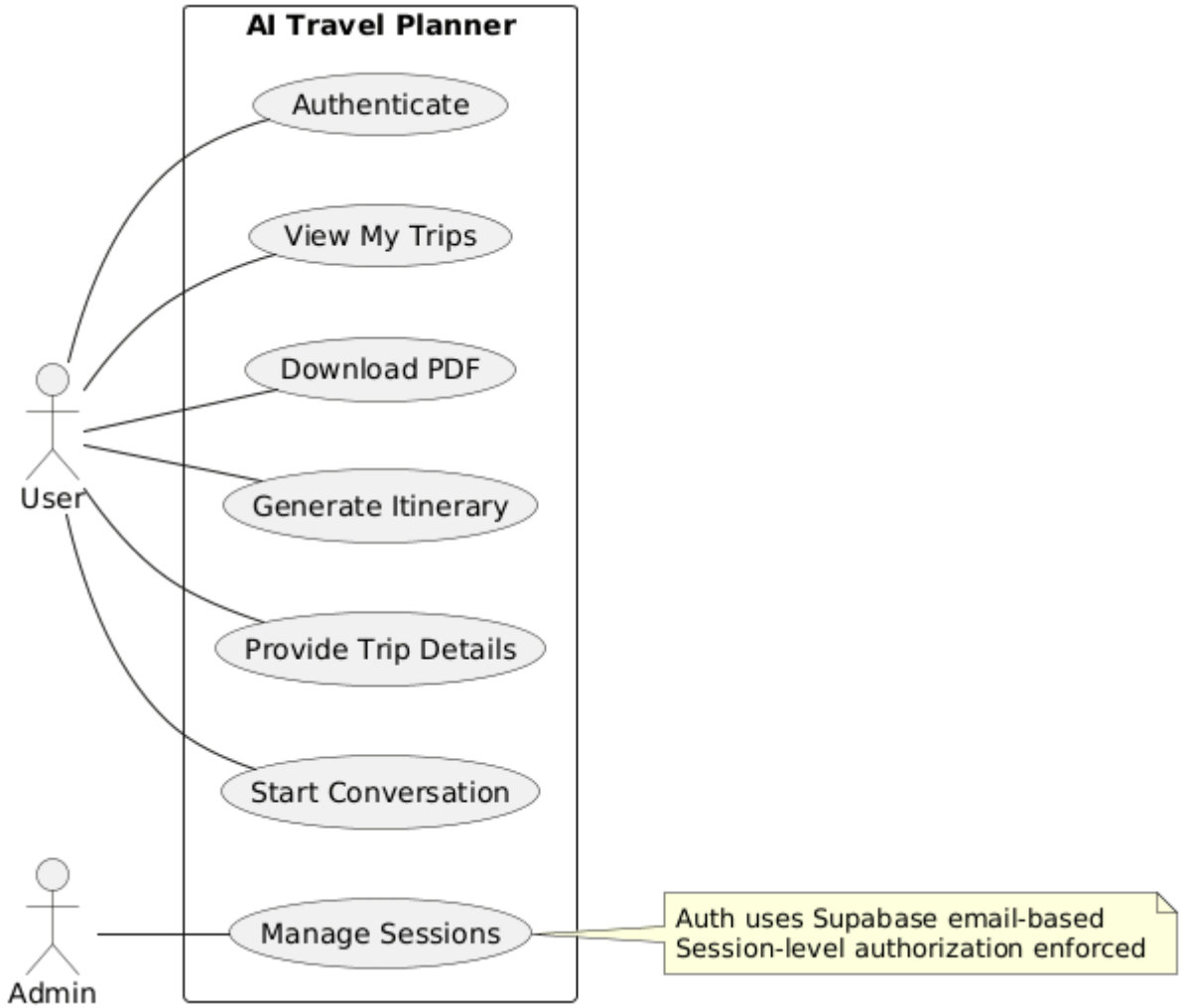
3.3.5 Sequence Diagram



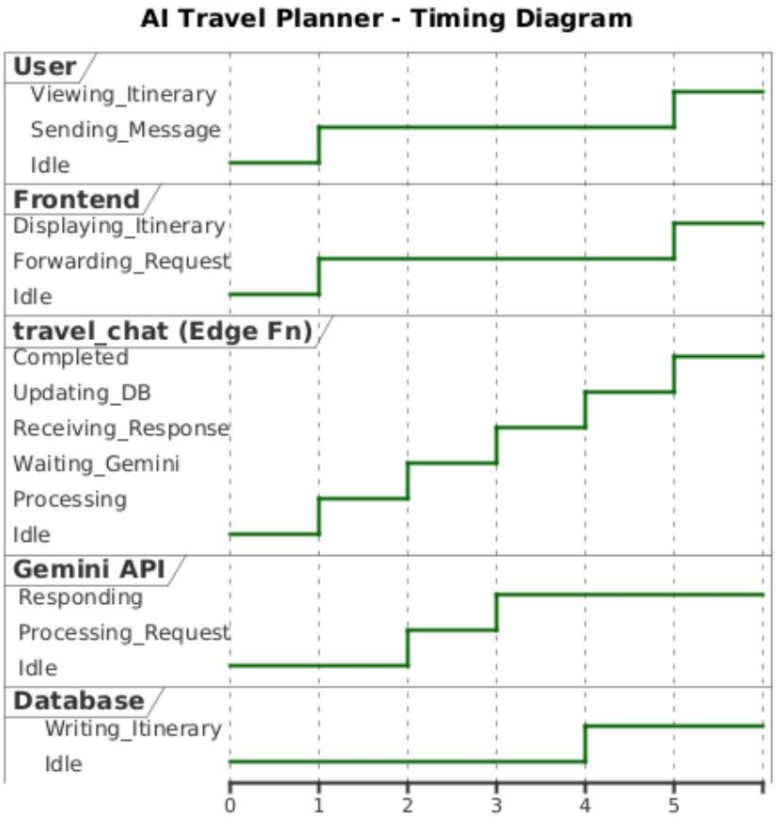
3.3.6 State Diagram



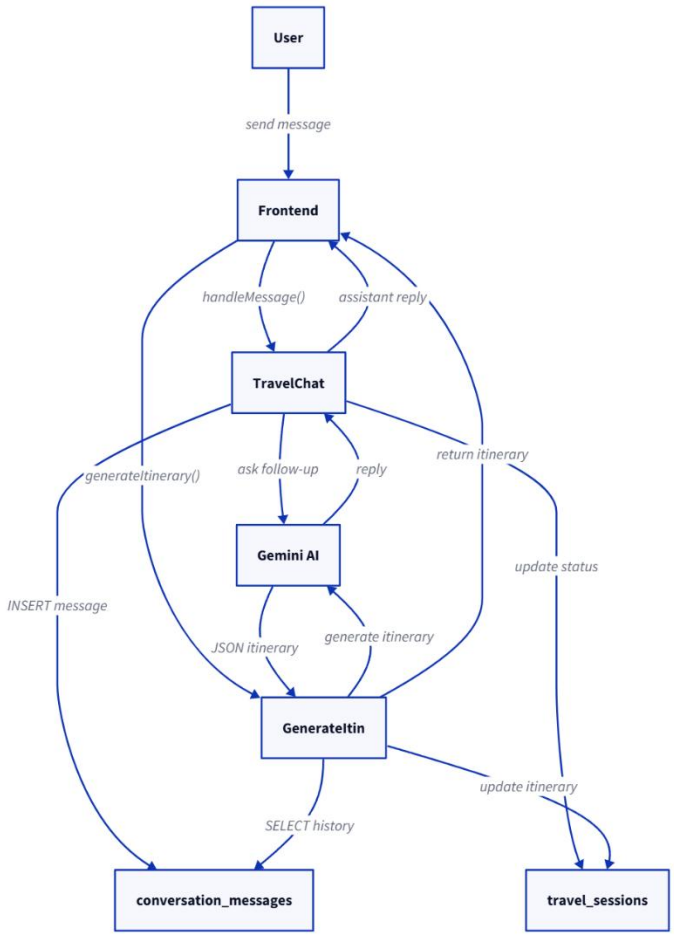
3.3.7 Usecase Diagram



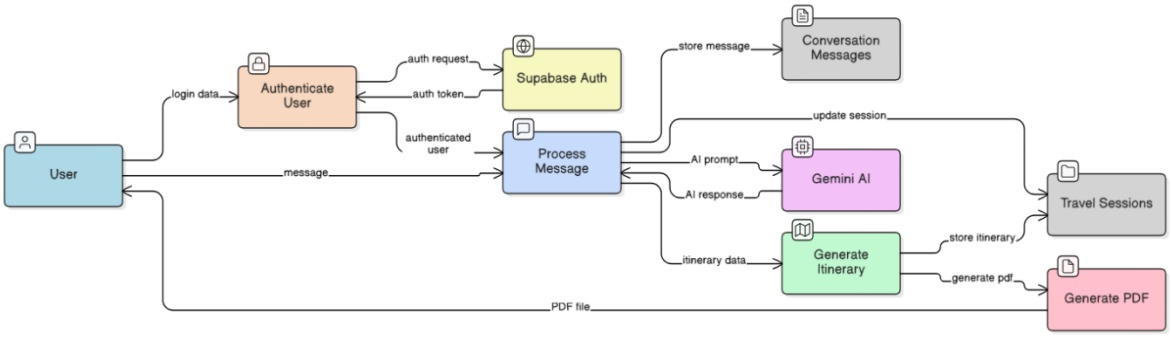
3.3.8 Timing Diagram



3.3.9 Collaboration Diagram

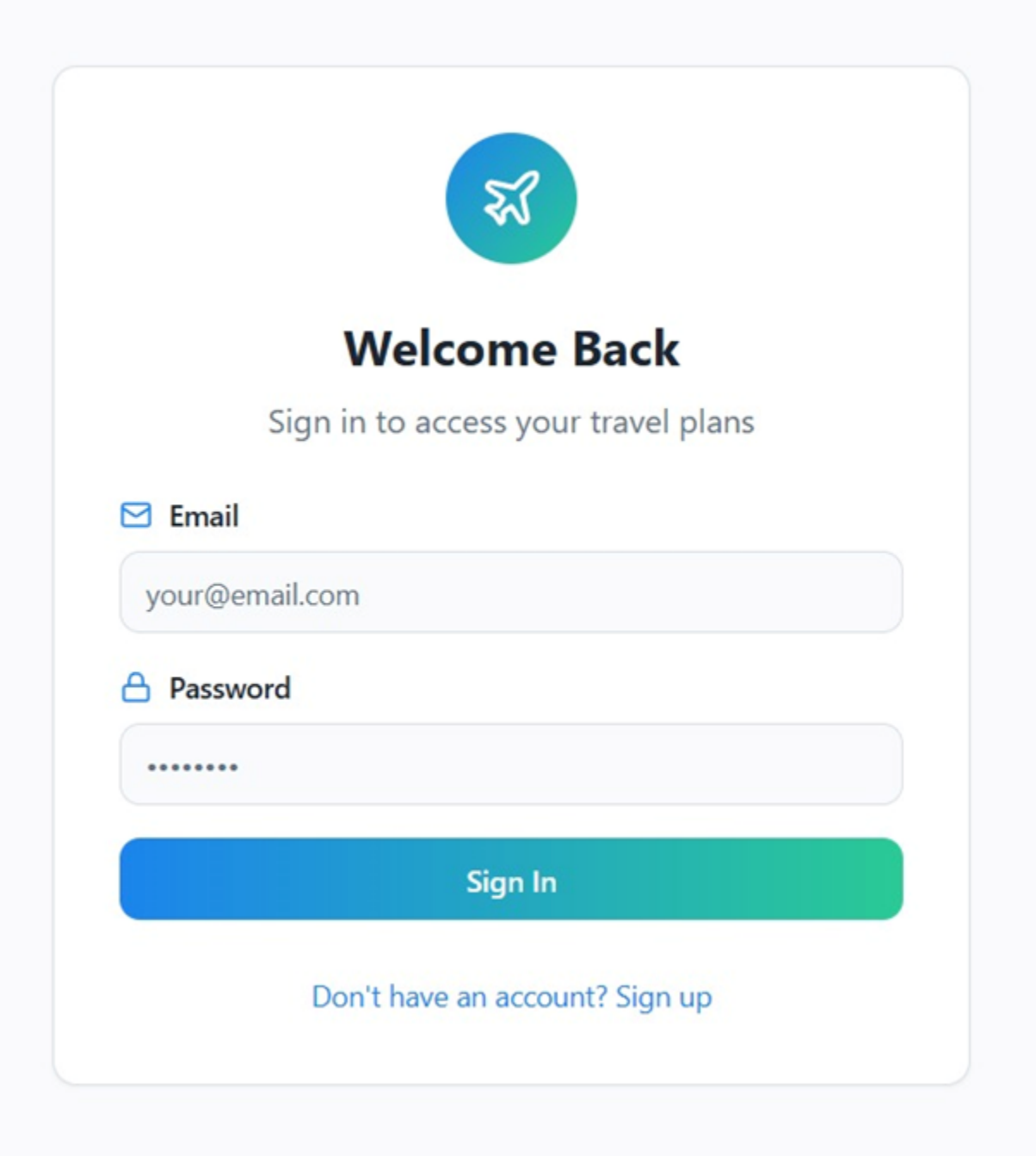


3.3.10 Data Flow Diagram

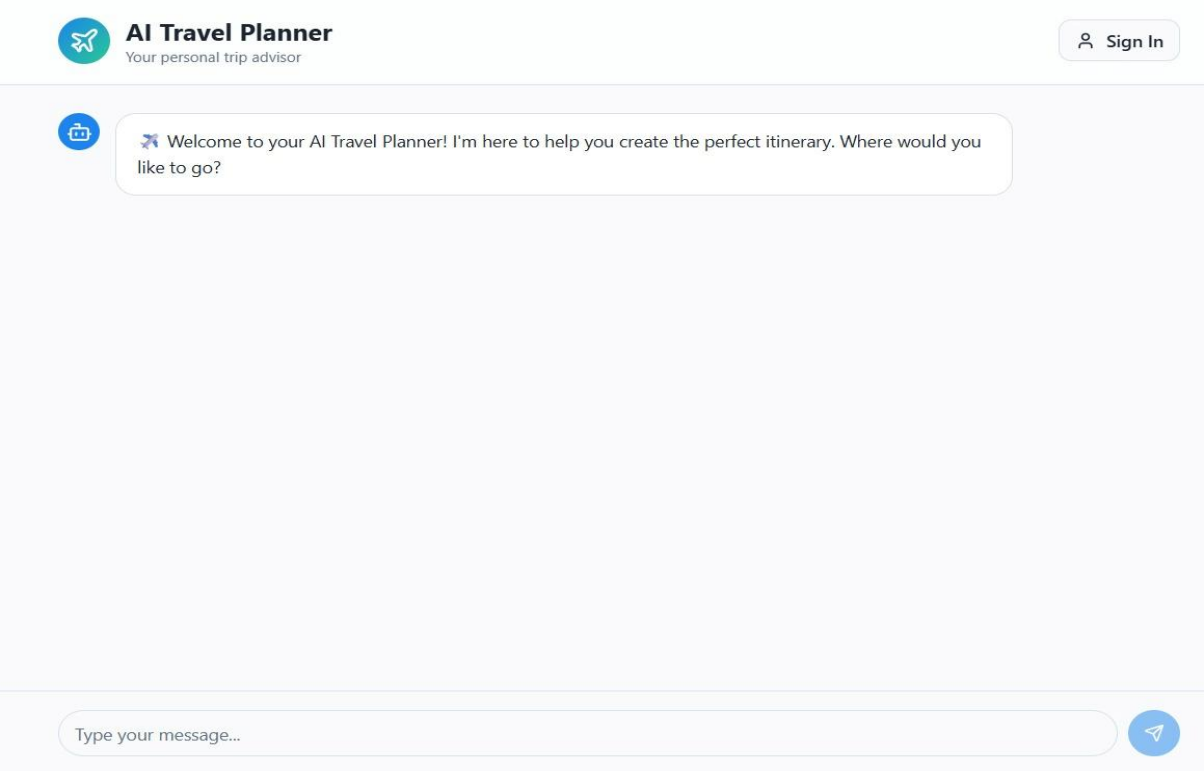


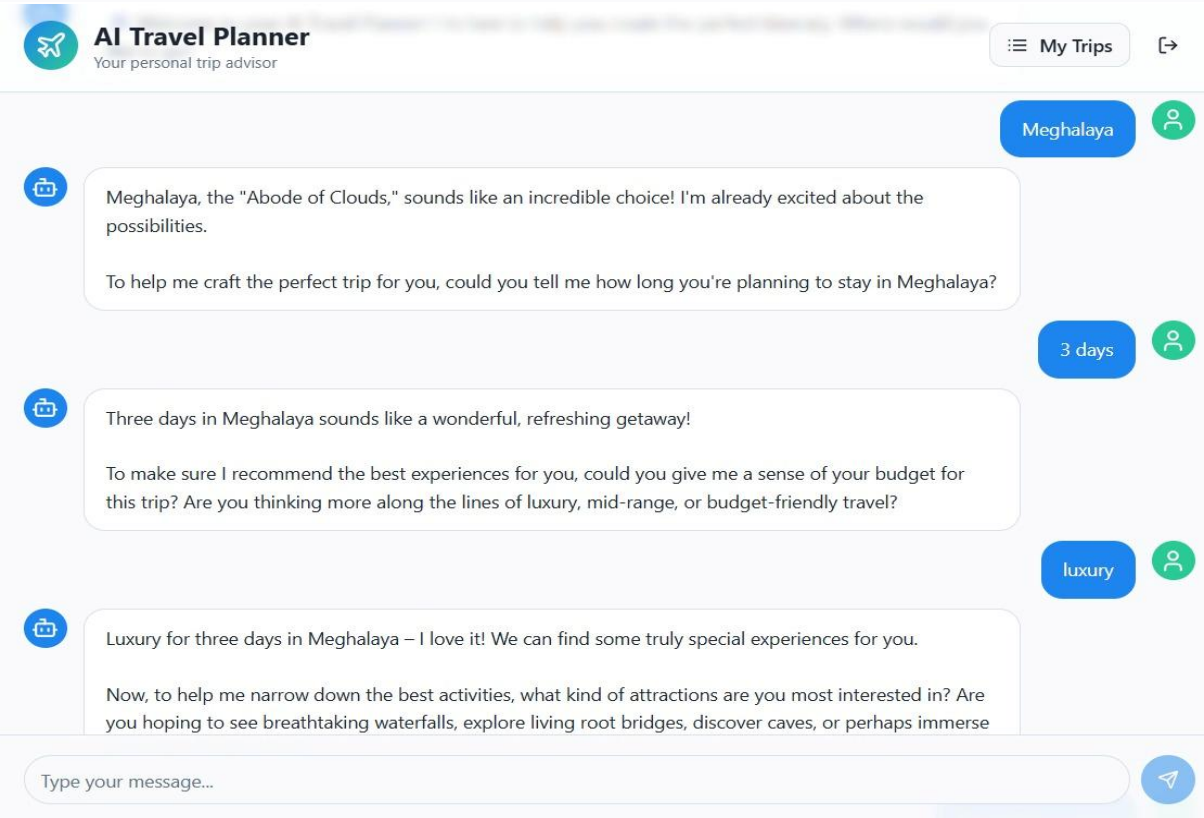
CHAPTER 4
IMPLEMENTATION

1. **Sign-in Module:**



2. Chatbot Onboarding:





CHAPTER 5
DISCUSSION

5.1 Discussion on the proposed activity

Our tourist app is set to redefine the way people travel by creating a fair, engaging, and sustainable platform for discovering local destinations. At its core, the app empowers users to highlight hidden gems through a democratic, user-voted system. This ensures that even the most underrated attractions receive the attention they deserve, based purely on merit and user feedback, rather than commercial promotions. In doing so, the app celebrates authenticity, encouraging travellers to go beyond typical tourist hotspots and immerse themselves in local culture.

Tourism, in its traditional sense, often overlooks smaller, less-commercialised destinations in favour of heavily advertised spots. This app changes that by levelling the playing field. Lesser-known attractions and businesses—such as family-owned restaurants, independent tour guides, and artisans—gain visibility, ensuring that tourists have access to unique and enriching experiences. At the same time, these local businesses benefit from increased footfall, leading to economic growth and the creation of new job opportunities. By connecting travellers directly with local service providers, the app creates a symbiotic relationship that benefits both parties.

The fairness of the app is driven by its meritocratic rating system, where reviews and recommendations are based on real user experiences. Unlike other platforms that often promote businesses based on paid ads, this system ensures transparency and authenticity. When users vote for destinations or leave reviews, their input helps shape the app's ecosystem, directing attention to places and businesses that genuinely excel. This not only builds trust among users but also incentivises businesses to maintain high standards, as their success is directly tied to customer satisfaction.

Beyond its focus on fairness and visibility, the app addresses a growing need for sustainability in tourism. Overcrowding in popular destinations often leads to environmental degradation and loss of cultural integrity. The app encourages travellers to distribute their visits more evenly, reducing pressure on overburdened locations. It also actively promotes responsible travel practices, such as using eco-friendly options, minimising waste, and supporting businesses that prioritise sustainability. By featuring "green" businesses and destinations, the app fosters a culture of environmental responsibility.

One of the standout features of the app is its focus on fostering community engagement. It goes beyond being a directory of destinations; it creates a platform where locals and travellers can interact meaningfully. Local residents can share insights about their

communities, recommend lesser-known attractions, and offer cultural tips. Travellers, in turn, gain a deeper appreciation of the places they visit, forging connections that go beyond a surface-level tourist experience. This exchange fosters mutual respect and understanding, making travel more rewarding for all involved.

The app also includes an educational element, enriching users' knowledge about the history and culture of the places they explore. Detailed descriptions of attractions, anecdotes from locals, and curated stories make the app more than just a travel guide. It becomes a tool for learning and cultural immersion, helping users engage with destinations on a deeper level. These features ensure that every journey is not just about sightseeing but about meaningful exploration.

To enhance user convenience, the app integrates real-time location-based services. Interactive maps help users find nearby attractions, local businesses, or even live events happening in the area. For instance, a traveller exploring a city can use the app to find a highly rated local café or a craft market within walking distance. This seamless integration of technology into the travel experience simplifies navigation and makes discovering new places effortless.

The app also provides a platform for promoting local events and festivals, further supporting cultural immersion. Travellers can discover events like traditional music performances, food festivals, or seasonal celebrations, giving them the chance to experience the local way of life. These events not only offer memorable experiences for tourists but also provide significant economic support to organisers and vendors.

Personalisation is another key feature of the app. By analysing user preferences, it can suggest destinations, activities, and businesses tailored to individual tastes. Whether someone is an adventure seeker, a foodie, or a history enthusiast, the app ensures that their travel experience is aligned with their interests. This level of personalisation creates a more enjoyable and fulfilling journey for each user.

One of the unique aspects of the app is its ability to promote economic inclusivity. Small and medium-sized businesses often struggle to compete with large corporations due to limited marketing budgets. The app addresses this issue by offering affordable advertising options that allow even the smallest vendors to reach a global audience. This creates a more equitable marketplace, where businesses of all sizes can thrive based on merit rather than resources.

The impact of the app is evident in its tangible outcomes. Lesser-known destinations will see a surge in visitor numbers, helping to distribute tourism more evenly and reduce overcrowding in popular spots. Local businesses will experience increased revenue and greater visibility, contributing to economic growth in their communities. Travellers, on the other hand, will benefit from authentic experiences and trustworthy recommendations, making their trips more satisfying and memorable.

The app also plays a vital role in raising awareness about the importance of sustainable and responsible tourism. By educating users about eco-friendly practices and encouraging them to support local economies, it contributes to a broader cultural shift in how people approach travel. This has long-term benefits, as it not only preserves the natural and cultural heritage of destinations but also fosters a sense of accountability among travellers.

Ultimately, the app is more than just a travel tool—it's a catalyst for change in the tourism industry. By creating a meritocratic, inclusive, and sustainable platform, it transforms the way people travel and engage with local communities. It promotes fairness, fosters cultural exchange, and encourages responsible behaviour, all while delivering immense value to both travellers and locals. In doing so, it lays the foundation for a tourism ecosystem that is equitable, enriching, and sustainable for years to come.

CHAPTER 6
ACTIVITY LOG

S. No.	Activity List	Date/s of activity completed	No. of hours spent
1	Identifying and Studying Research Papers	20/09/2024	8
2	Identifying and Evaluating Existing Solutions	21/09/2024	8
3	Identification of Key Stakeholders to Optimise Offering	27/09/2024	5
4	Identification of Modules	12/10/2024	8
5	Feasibility Study of Proposed Modules	16/11/2024	8
6	Identifying Technical Requirements for the Modules	22/11/2024	9
7	Estimation of Project Completion Timeline	23/11/2024	5
8	Preparation of the Presentation	30/11/2024	4
9	Preparation of the Report	01/12/2024	5

S. No.	Activity List	Date/s of activity completed	No. of hours spent
10	Development: Sign In Module	01/02/2025	8
11	Development: Onboarding Chatbot (I)	19/03/2025	8
12	Development: Onboarding Chatbot (II)	26/03/2025	4
13	Integration – Core Modules	04/05/2025	5
14	Integration – User Interface Layer	11/05/2025	5
15	Initial QA	19/06/2025	6
16	Bug Fixing	26/06/2025	4
17	Stakeholder Review Session	04/08/2025	6
18	Feedback Implementation	19/09/2025	8
19	Technical Documentation (Sign In & Onboarding Modules)	05/11/2025	6